National Financial Planning
for
Natural Hazard Risk Mitigation

Sri Lanka



Introduction

- Disasters present a broad range of human, social, financial, economic and environmental impacts, with potentially long-lasting, multi-generational effects.
- The financial management of these impacts is a key challenge for individuals and governments in developing countries.
- Addressing natural hazard-related issues involves a multi-faceted approach.
  - Financial planning
  - Risk management
  - Policy development

Risk management methodology



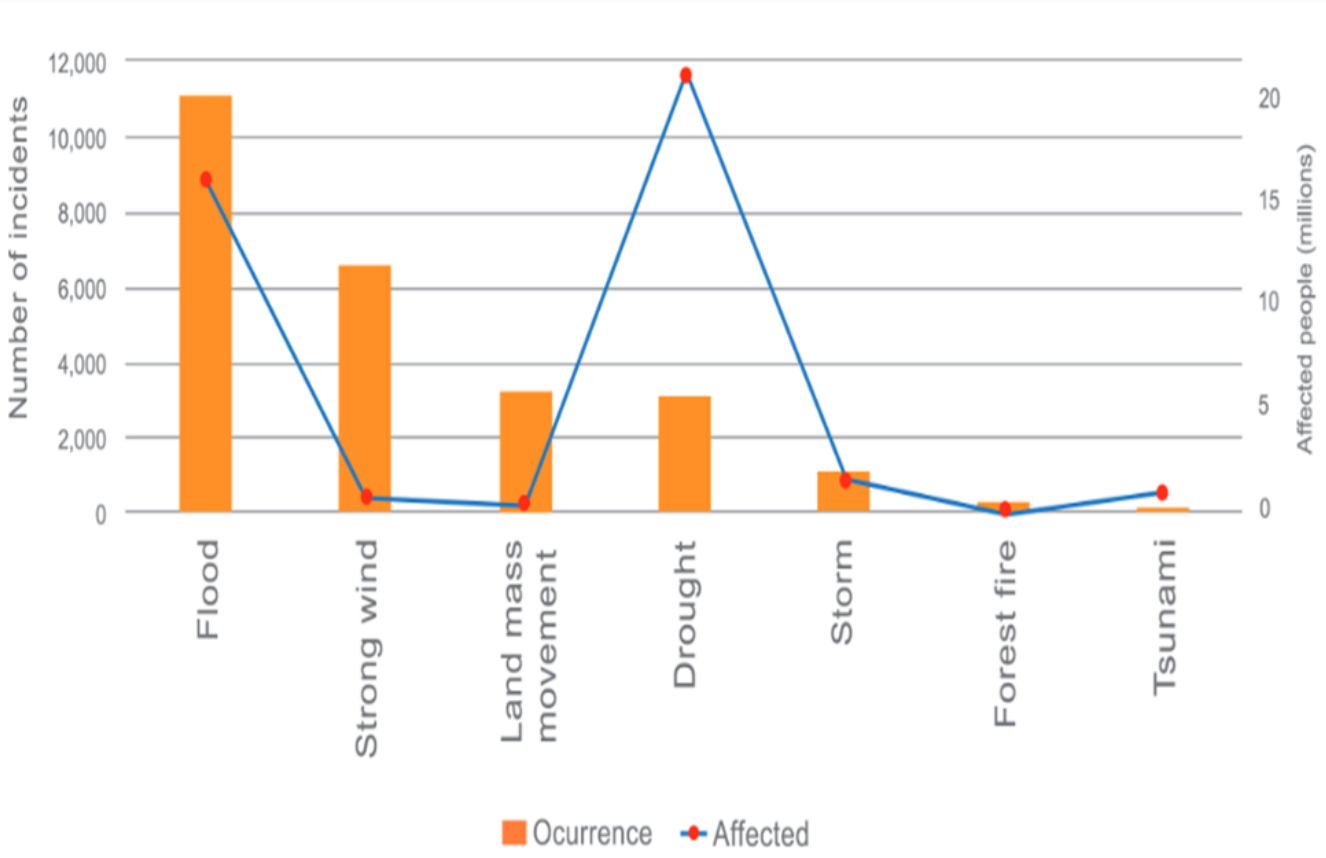
Developing an inventory of assets → Modelling natuaral hazards → Modelling risk financing

Sri Lankan context

Risk potential

- In accordance with the ND-GAIN indices (Notre Dame Global Adaptation Initiative) Sri Lanka is one of the countries with a high risk to disasters.
- Accordingly, the country faces significant challenges in terms of financing losses and damages from natural disasters.
- The severity of hazards is expected to be exacerbated due to the climate change

Most frequent natural hazards from 1965 to 2020



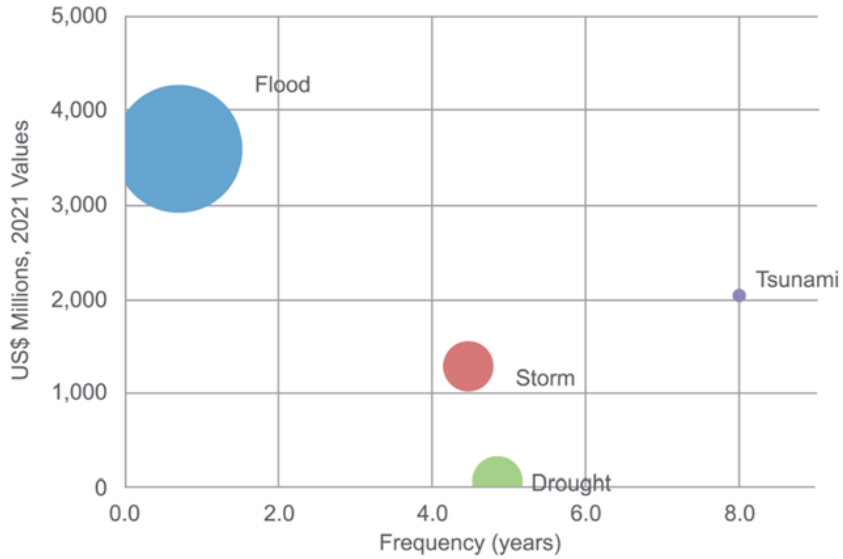
ND-GAIN Index=Readiness Score−Vulnerability Score(higher the score higher the adaptability)

Where Vulnerability Score=Exposure+Sensitivity−Adaptive Capacity

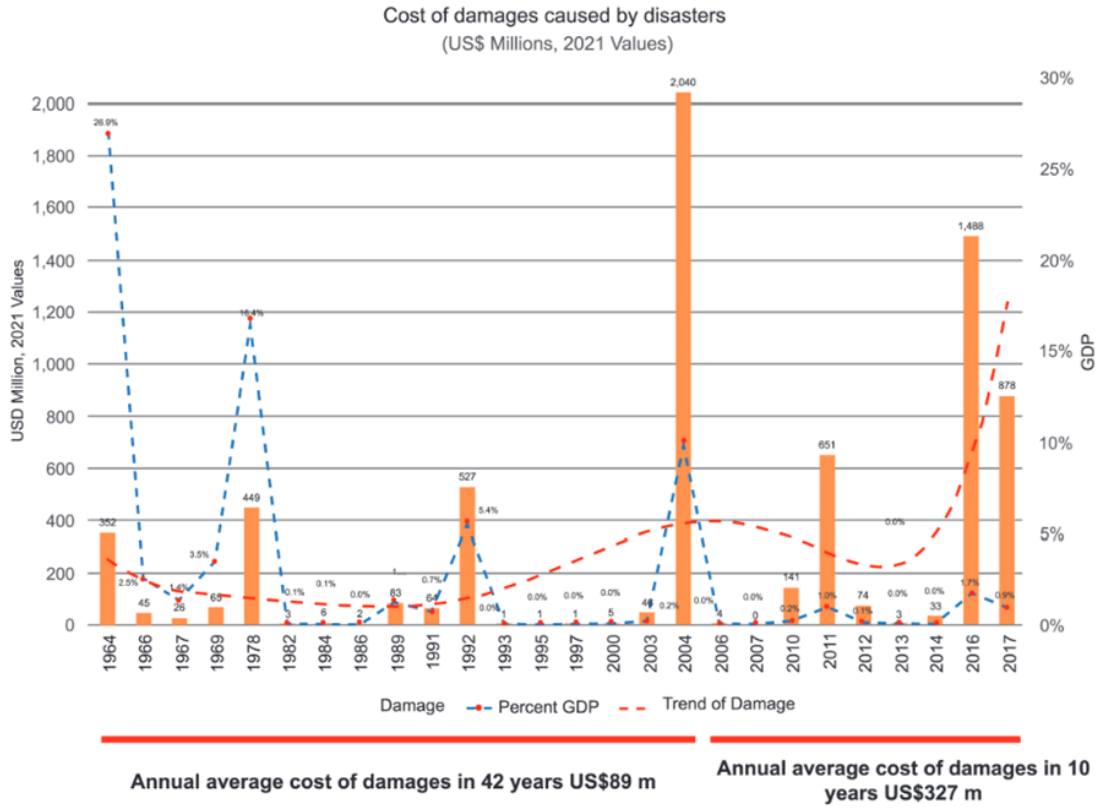
Readiness Score=Economic Readiness+Governance Readiness+Social Readiness

Cost of damage

- The key driver for explaining the cost of damages historically are floodings.
- For the period assessed, approximately 50% of the damages arose from flooding events.
- Tsunami accounted for 30% and the rest (20%) were due to other natural hazards



Frequency and costs by type of hazards from 1964 to 2021



Costs by type of hazards from 1964 to 2021

Financial spending for Disaster Management

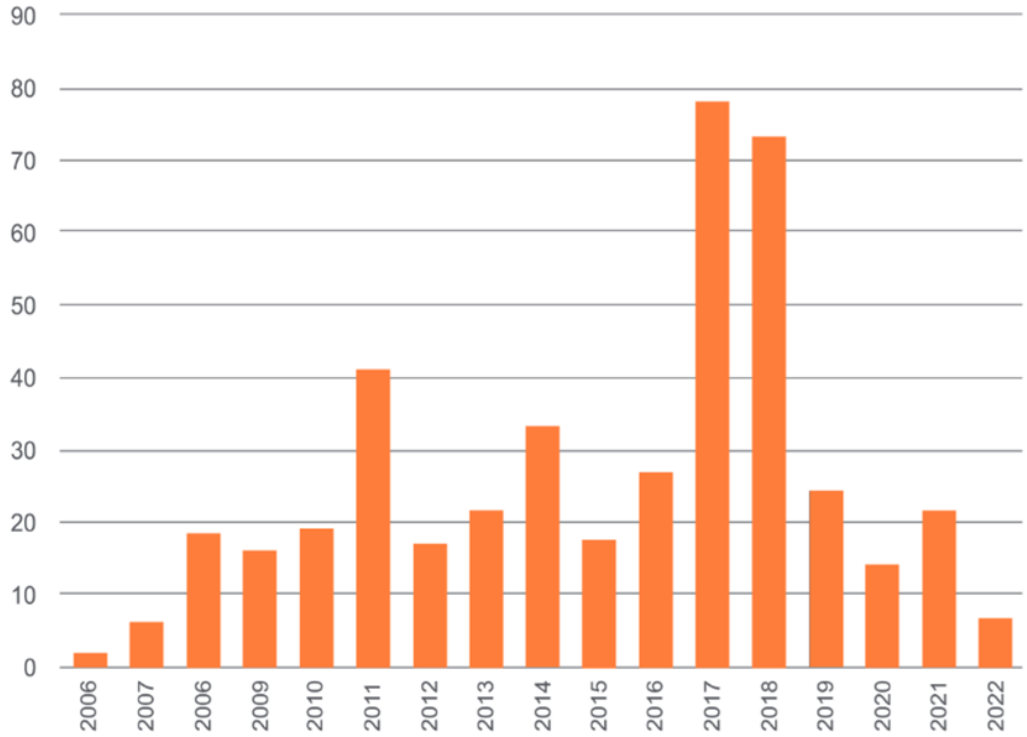
100% ex-post financing

Current sources of financing

- National budget
  - ✓ Individual budget allocation to ministries

- The former national natural disaster insurance scheme
  - ✓ to compensates the cost of damages to the population through a universal insurance scheme for individuals and small businesses, managed by the National Insurance Trust Fund

- International donor community and international support



Government fund allocation for disaster risk management

(Source - Desinventar Database)

Suggested financial means

- Explicit annual budget allocation for disasters
- Disaster funds
- Contingent credits
  - ✓ An ex-ante financial instrument
  - ✓ An agreement between a financial institution(multilateral) and a government
  - ✓ Appropriate to finance losses with a low probability of occurrence

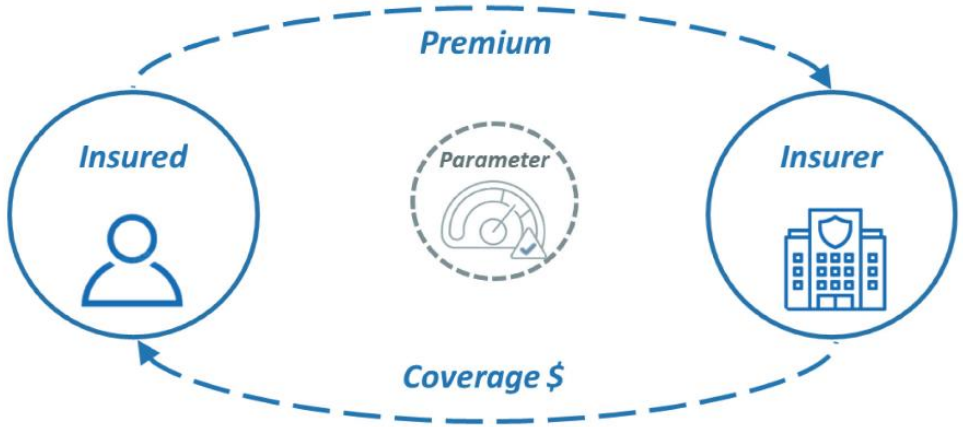
Risk retention

- Traditional Insurance
- Parametric Insurance
  - ✓ a contract whereby the insurer undertakes to compensate, when the previously agreed parameters are met.
  - ✓ is relevant in countries with lack of data quality and quantity.
  - ✓ is based on the probability of occurrence of an event and not the loss it could cause.
  - ✓ This insurance is measured and/or calculated by a third party using a predetermined methodology.
- Capital market: Catastrophe bonds

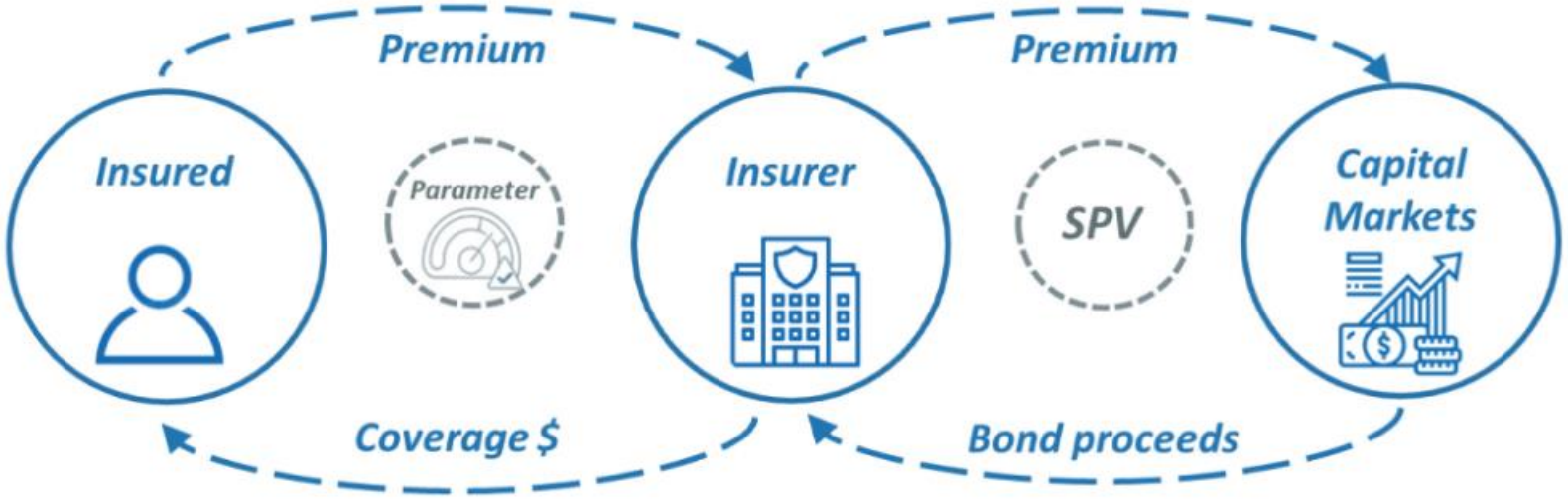
Risk transfering



Traditional insurance



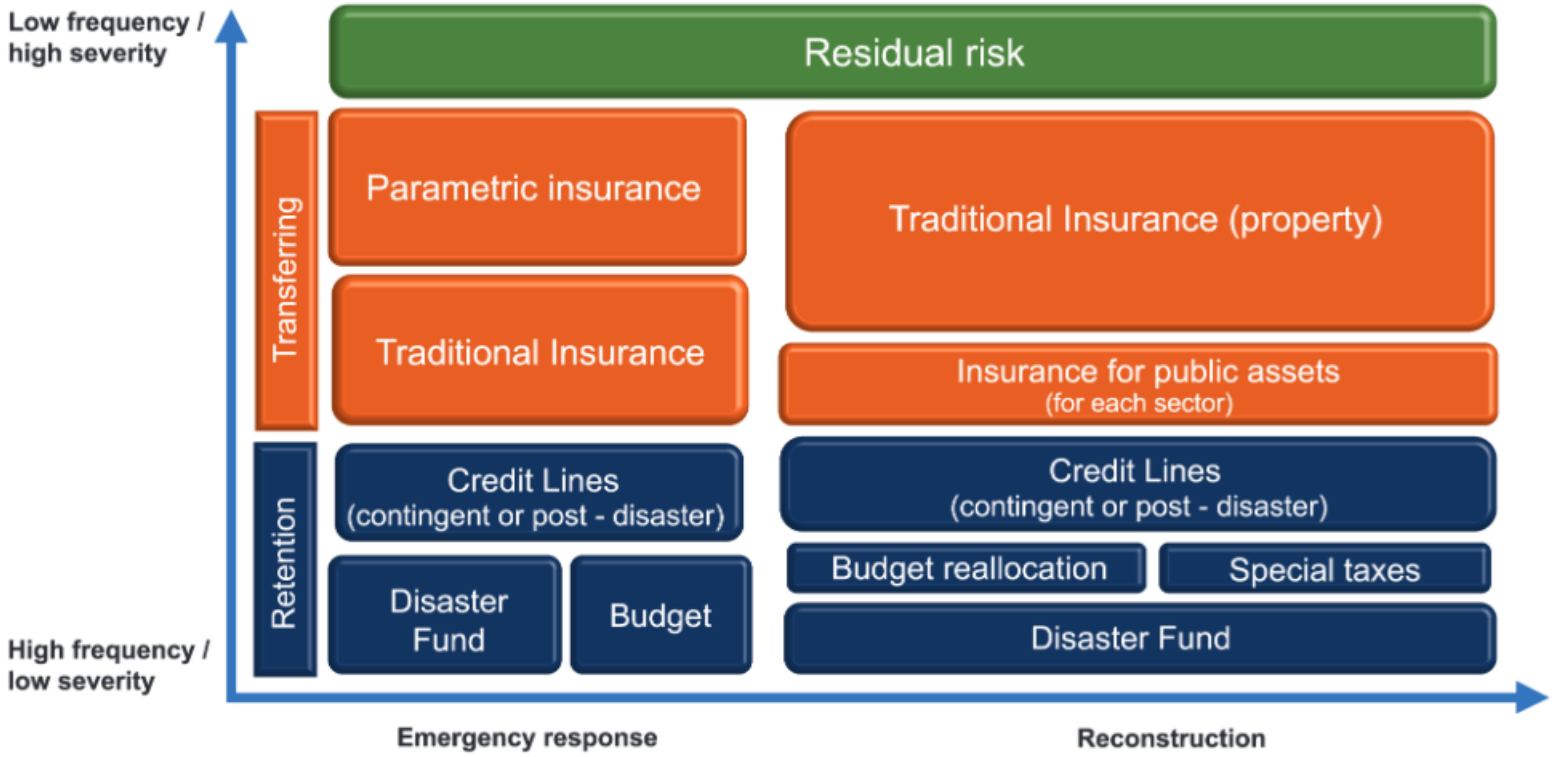
Parametric insurance



Catastrophe bonds

(SPV= Special Purpose Vehicle)
The SPV is a separate legal entity that enters into an insurance contract with an insurance company to issue the catastrophe
bond securities in exchange for payment from investors who purchase the security.

Risk layering methodology in risk financing



Source-www.undp.org

Recommendations

- An ex-ante disaster risk financing strategy can provide financing for specific sectors.
  - This pre-arranges funding mechanisms to finance disaster response and recovery efforts in advance of a disaster occurrence.
  - Reduces the financial burden of disasters on public budgets and ensuring a timely and effective response to disasters.

Challenges

- There is no institutionalized mechanism to systematically and periodically gather information on the damages and losses that are associated with disasters.
- There is also a lack of data breakdown by sector, at the subnational and local levels.

Data sources

- Disaster Management Centre and UNDP(2023), Financial planning for disaster preparedness: Irrigation Sector in Sri Lanka

- Disaster Management Centre Profile.

  http://www.dmc.gov.lk

- Organisation for Economic Co-operation and Development (OECD), (2015). Disaster Risk Financing: A global survey of practices and challenges. OECD Publishing, Paris.

  http://dx.doi.org/10.1787/9789264234246-en